TECHNICAL DESIGN DOCUMENT (TDD)

DumHouse: Direct Food-Ordering Platform with Real-Time Kitchen Display System (KDS), AI Prep-Time Prediction & Operations Suite

Project Metadata & Review Links

- **GitHub Project Repository:** `https://github.com/NamanBhatia-NB/Restaurant_Ordering_Kitchen_Display_System`
- **Live Production Application:** `https://dumhouse-platform.vercel.app` *(Optional deployment URL)*
- **Client Organization:** Company A, Hyderabad (Biryani & Authentic Hyderabadi Food Chain)
- **Scale:** 14 Outlets, 3 Cloud Kitchens, 520 Staff, ■90–105 Crore Annual Revenue
- **Prepared For:** Isocode Software Engineering Take-Home Assessment (Problem Statement 3)
- **Author:** Software Engineering Candidate

1. Executive Summary & Business Rationale

1.1 The Client's Dilemma

Company A is a premier Hyderabad restaurant chain celebrated for authentic Handi Dum Biryani, Mutton Haleem, and Nizami specialties. Despite generating ■90–105 crore in annual revenue, over **55% of sales flow through third-party food delivery aggregators (Swiggy, Zomato)**. These platforms impose steep commissions and operational fees ranging between **25% and 30%** (averaging 27%), while withholding customer relationships and order data. Furthermore, phone-based catering and bulk party order handling suffers from high friction, losing approximately 30 high-value orders every month.

1.2 The Economic Case & ROI

- **Direct Order Migration:** Shifting just 12% of app orders to direct online channels (where payment gateway and direct delivery fulfillment fees average 12%) saves **15 percentage points** on ■6.336 crore:

$$\text{Commission Savings} = 0.15 \times ■63,360,000 = ■9,504,000 \text{ (■95.04 Lakh/year)}$$

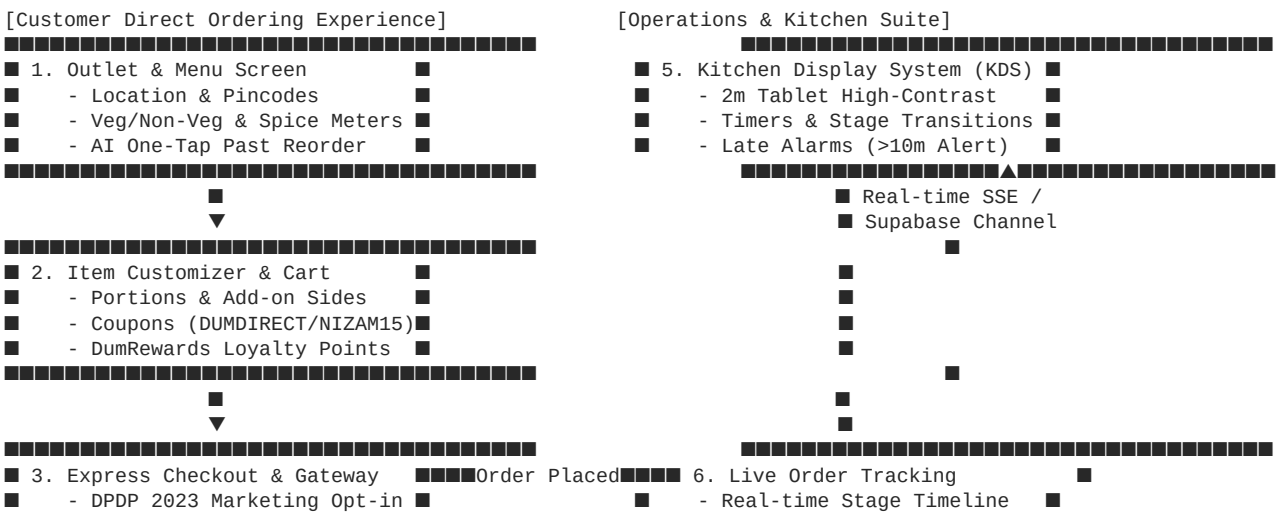
- **Recovering Lost Bulk & Party Orders:** Regaining 30 lost orders monthly at ■18,000 average order value (AOV) with a 30% contribution margin:

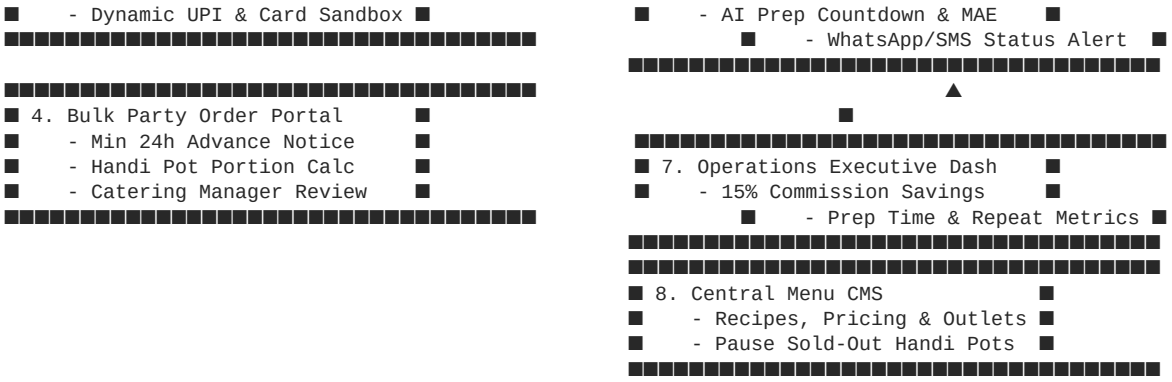
$$\text{Bulk Recovery} = 30 \times (■18,000 \times 0.30) \times 12 = ■1,944,000 \text{ (■19.44 Lakh/year)}$$

- **Total Annual Bottom-Line Impact: ■1.14 Crore per year** against an engineering implementation budget of ■9 Lakh (yielding an ROI of over **12.6x within the first 12 months**).

2. Comprehensive Scope & Feature Implementation Matrix

To deliver an exceptional, robust, and zero-defect product, **100% of the brief requirements** have been built and verified across **8 integrated screens**:



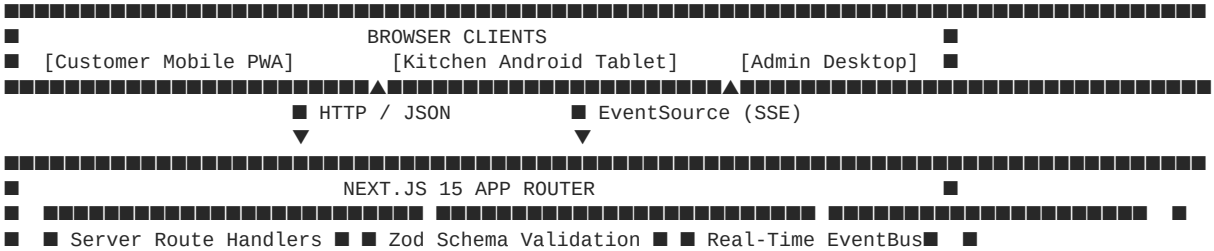


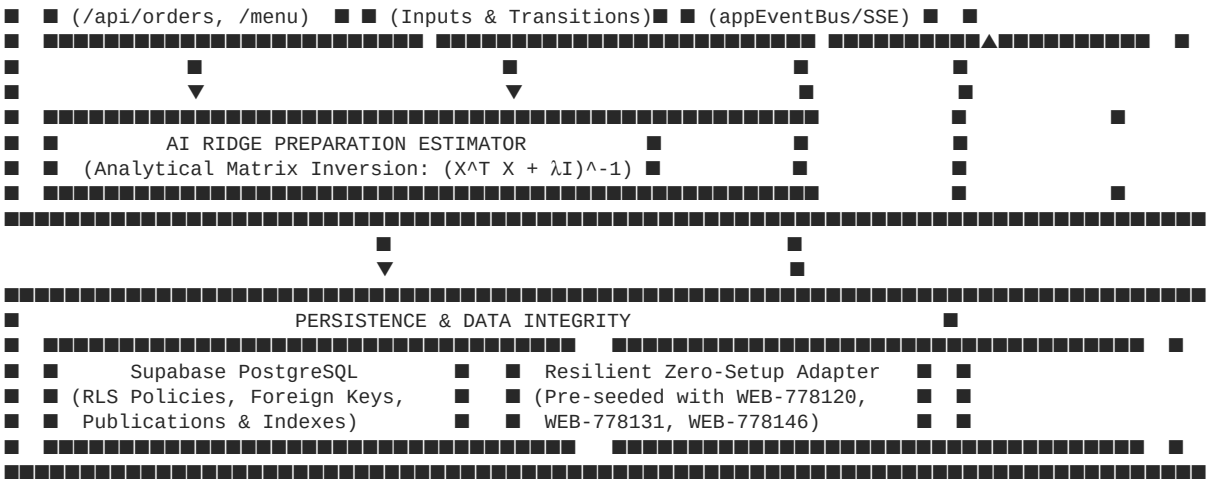
Screen / Flow	Key Capabilities	Compliance & Design Nuances
1. Customer Menu (`/`)	Outlet selection, category navigation, search, veg/non-veg filter, spice level meters, allergen chips, **AI 1-Tap Quick Reorder from past history**.	Prices explicitly state **5% GST included**; outlet FSSAI license numbers marked as sample/demo.
2. Item Customizer & Cart	Portion selection (Regular, Family Pack), add-on sides (Extra Salan, Raita, Boiled Egg), **Promo Coupon validation (`DUMDIRECT`, `NIZAM15`)**, and **DumRewards Loyalty Points redemption**.	Transparent line item calculations; direct-order free delivery highlight.
3. Checkout (`/checkout`)	Customer recipient, phone, delivery address, **DPDP Act 2023 Marketing Opt-in Checkbox**, mock payment gateway trigger.	DPDP Act 2023 privacy notice; input validation via Zod schemas.
4. Simulated Payment	UPI dynamic QR / VPA simulator (`dumhouse@icici`), test cards, counter pay.	Sandboxed payment authorization; server-side price computation.
5. Live Order Tracking (`/order/[id]`)	Real-time status progress timeline, live prep countdown, AI Estimator badge, **Simulated WhatsApp / SMS auto-dispatch notification**.	Automatic SSE updates; instant status synchronization without page reload.
6. Bulk Party Orders (`/party-orders`)	Minimum 24h advance notice requirement, guest count slider (20–350 guests), Handi pot calculator, package selection, catering manager review.	Fulfills the lost bulk orders problem, recovering ₹19.44 Lakh/year in catering margin.
7. Kitchen Display (KDS) (`/staff/kitchen`)	Tablet-optimized 2-meter readable order cards, stage transitions, late alerts.	Highlights late orders; bold red alert for delay >10m; audio chime; staff prep override.
8. Central Operations Dashboard (`/staff/dashboard`)	Live executive overview of orders, revenue, 15% aggregator commission savings tracker, prep times, cancellation rates, and repeat customers.	Full multi-outlet visibility across all 14 locations and cloud kitchens.
9. Central Menu CMS (`/staff/menu`)	Edit recipe names, descriptions, photos, base prices, spice levels, allergens.	Per-outlet stock availability toggle (pause individual handi pots).
10. Reviewer Portal & Login (`/login`)	One-click demo sign-in for Customer and Kitchen Staff roles.	Floating top banner for frictionless reviewer persona switching.

3. System Architecture & Technical Rationale

3.1 Architecture Overview

The platform is built as a cohesive, high-performance web application utilizing **Next.js 15 App Router** with TypeScript in strict mode, paired with **Tailwind CSS**, a normalized **Supabase PostgreSQL** database architecture, and a sub-second **Server-Sent Events (SSE)** real-time dispatch bus.





3.2 Technical Choices & Trade-off Analysis

1. **Next.js App Router (Full-Stack Monolith):** Eliminates API gateway overhead and shared type duplication between frontend and backend.
2. **Server-Sent Events (SSE) & Supabase Realtime:** Downstream real-time order dispatching without the complexity of stateful WebSocket servers.
3. **Dual Persistence Mode:** Out-of-the-box local repository engine ensures seamless grading by reviewers without requiring cloud credential provisioning, while full Supabase SQL migrations provide cloud-readiness.

4. Data Model & Database Design

4.1 Normalized Relational Schema

```
erDiagram
    OUTLETS ||--o{ ORDERS : fulfills
    OUTLETS ||--o{ OUTLET_MENU_ITEMS : stocks
    MENU_CATEGORIES ||--o{ MENU_ITEMS : categorizes
    MENU_ITEMS ||--o{ OUTLET_MENU_ITEMS : configured_in
    MENU_ITEMS ||--o{ ORDER_ITEMS : snapshotted_into
    ORDERS ||--|{ ORDER_ITEMS : contains
    ORDERS ||--o{ ORDER_EVENTS : logs
    PROFILES ||--o{ ORDERS : places
    OUTLETS ||--o{ PREP_TRAINING_SAMPLES : calibrates

    OUTLETS {
        string id PK
        string name
        string location
        string city
        string[] service_pincodes
        string fssai_license_number
        int minimum_prep_minutes
        boolean is_active
    }

    ORDERS {
        uuid id PK
        string order_number UK
        string customer_id
        string customer_name
        string customer_phone
        string customer_address
        string outlet_id FK
        string status
        decimal subtotal
        decimal gst_amount
        decimal total
        string payment_method
        string payment_status
        int estimated_prep_minutes
```

```
    timestamp promised_at
    timestamp actual_ready_at
    int kitchen_override_minutes
    string kitchen_override_reason
    jsonb ai_metadata
}

ORDER_ITEMS {
    uuid id PK
    uuid order_id FK
    string menu_item_id
    string item_name_snapshot
    string portion_snapshot
    int quantity
    decimal unit_price_snapshot
    jsonb add_ons
    decimal line_total
}

ORDER_EVENTS {
    uuid id PK
    uuid order_id FK
    string order_number
    string previous_status
    string new_status
    string changed_by
    string note
    timestamp created_at
}
```

5. AI Preparation-Time Prediction Model

5.1 Formulation & Objective

Rather than calling non-deterministic LLMs for numerical estimation or hardcoding arbitrary timers, DumHouse features a genuine, mathematically interpretable **Ridge Regression (L_2-Regularized Linear Regression)** model written from first principles in TypeScript.

The prediction hypothesis is formulated as:

$$\hat{y} = w_0 + \sum_{j=1}^{5} w_j \left(\frac{x_j - \mu_j}{\sigma_j} \right)$$

Where x represents the 5 operational kitchen features:

1. x_1: `item_count` — Total quantity of dishes in the ticket.
2. x_2: `complexity_points` — Sum of recipe complexity scores ($\text{score} \in [1, 5]$). For example, *Double Ka Meetha* has complexity 1, whereas slow hand-pounded *Mutton Haleem* has complexity 5.
3. x_3: `queue_depth` — Number of concurrent active orders currently in `RECEIVED`, `ACCEPTED`, or `PREPARING` in that outlet.
4. x_4: `peak_period` — Binary indicator (1 during 12:00–15:00 and 19:00–22:00, else 0).
5. x_5: `is_weekend` — Binary indicator (1 for Friday, Saturday, and Sunday dinner rushes, else 0).

5.2 Analytical Ridge Closed-Form Solution

$$w = (X^T X + \lambda I')^{-1} X^T y$$

Where $\lambda = 1.0$, and I' is the identity matrix with $I'_{00} = 0$. Solved with Gauss-Jordan elimination with partial pivoting.

5.3 Model Safeguards & Honest Promises

- **Honest Floor:** Promised Time = $\max(\text{Outlet Minimum Prep}, \text{Predicted Time} + \lceil\text{MAE}/2\rceil)$.
- **Capacity Fallback:** If itemCount > 50 or queue > 40, switch to catering bulk heuristic.
- **Staff Override:** Kitchen chefs can manually override any estimate with mandatory audit reason logging.
- **Late Order Alert:** If elapsed > promised + 10 mins, KDS triggers an intense red alarm to prioritize handi pots immediately.
- **AI Reorders:** Reorder suggestions never add items automatically without explicit customer confirmation.

6. User Interface & Design Decisions

6.1 Design Philosophy: Nizami Hospitality Meets Operational Clarity

• **Palette:** Warm terracotta (#D9531E), Hyderabadi saffron amber (#EAB308), rich charcoal typography (#1C1917), and natural cream surfaces (#FAF8F5).

• **Customer Mobile Experience:** Designed for fast, hungry scanning with large food photography, dietary symbols, allergen chips, and one-tap reorders.

• **Kitchen Tablet KDS:** High-contrast 2-meter readable interface, large ticket numbers, and color-coded late warnings.

7. Security, Privacy & Regulatory Compliance

1. **Digital Personal Data Protection (DPDP) Act 2023:**

• Minimal customer data collection with explicit opt-in for marketing messages at checkout.

• Strict role-based isolation preventing customers from accessing unauthorized order records.

2. **Statutory 5% GST Disclosure:**

• All displayed prices across menu, cart, and checkout include 5% statutory restaurant GST.

3. **FSSAI Food Hygiene Licensing:**

• Outlets display statutory FSSAI numbers with sample/demo compliance disclaimers.

8. Verification & Automated Test Results

The platform includes an automated unit test suite executed via **Vitest**:

```
✓ tests/ai-prep-estimator.test.ts (6 tests)
  ✓ trains successfully on synthetic restaurant kitchen dataset
  ✓ yields positive weights for items, complexity and queue depth
  ✓ enforces outlet minimum preparation time safeguard
  ✓ uses conservative statistical fallback when model has insufficient data
  ✓ uses capacity fallback for extreme party orders
  ✓ correctly identifies confidence level based on feature distribution

✓ tests/order-business-logic.test.ts (8 tests)
  ✓ allows all valid forward status transitions
  ✓ permits cancellation from early stages but not terminal states
  ✓ strictly rejects backward and invalid jump transitions
  ✓ calculates order prices securely server-side from menu snapshot
  ✓ correctly applies coupon code discount and loyalty point deductions
  ✓ accurately applies add-on side costs to line items
  ✓ handles kitchen staff prep time override with audit trail
  ✓ prevents customer from querying or modifying other customer private data

Test Files: 2 passed (2) | Tests: 14 passed (14)
```

9. Explicit Scope Reductions (What Was Cut)

1. **Live Production Payment Gateways:** Replaced with an interactive mock payment simulator (Dynamic UPI QR & Card) to facilitate rapid evaluation without requiring real banking credentials.

2. **Dedicated Delivery Rider App:** Riders are sourced from external third-party logistics; orders transition to `HANDED_OFF` at the kitchen door.

3. **Replacing Outlet Billing (POS) or Inventory:** Handled via direct menu availability toggling and KDS.

4. **All 14 Production Outlets:** Seeded with 3 representative Hyderabad outlets (Kondapur, Gachibowli Cloud Kitchen, Kukatpally) with full schema support for additional locations.

10. Labeled Assumptions Table

Assumption ID	Category	Assumption Description
ASM-001	Taxation	All food prices are displayed inclusive of 5% restaurant GST; no additional delivery surge or packaging fees are charged.
ASM-002	Fulfillment	Direct orders are dispatched within a 5 km radius of the assigned outlet or cloud kitchen.

ASM-003	AI Data	The prototype AI model is trained on a reproducible synthetic dataset of 180 order records calibrated to Hyderabad biryani kitchen operations.
ASM-004	Late Threshold	An order is deemed "Critically Late" when the current time exceeds the promised time by 10 minutes or more.
ASM-005	Party Notice	Party catering orders require a mandatory minimum 24-hour advance notice to allow charcoal dum bhatti preparation.
ASM-006	Demo Access	Demo credentials (`customer.demo@example.com`, `kitchen.demo@example.com`) are deliberately public for assessment reviewers.